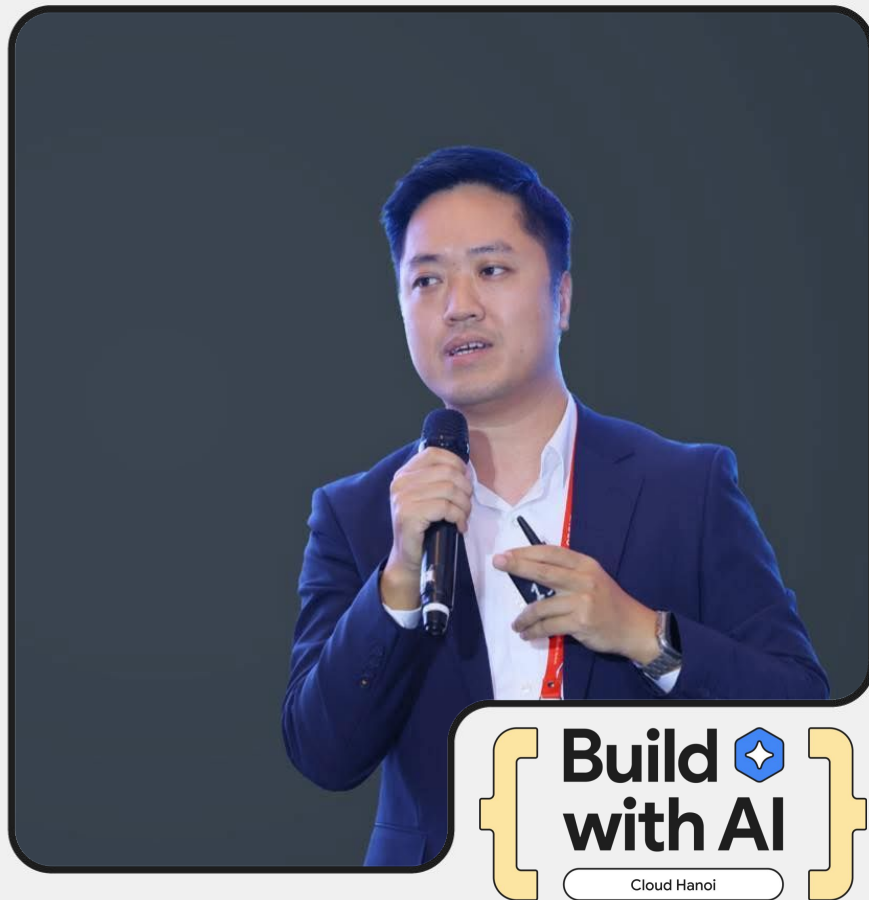


Gemma4 - Open Source AI, Redefined.

Thang Nguyen - Head of
Development





Thang Nguyen

Head of Development,
Sai Gon - Hanoi Bank
Founder @ Mi AI
(miai.vn)



miai.vn

Học AI theo cách mì ăn liền!

Let's **connect!**



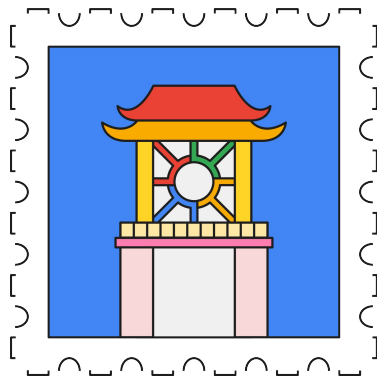
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Agenda

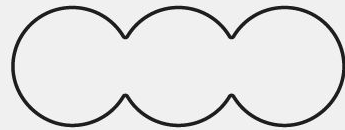
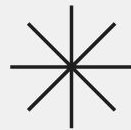
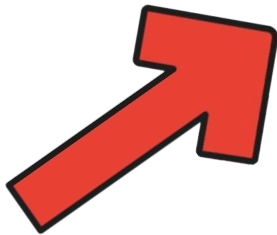
- 1 About Gemma4
- 2 Key features
- 3 Ecosystem & Tools
- 4 Demo
- 5 Q&A



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About Gemma4



{ Build  with AI }

Today event is

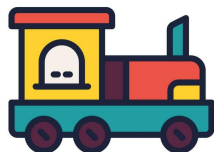
Build with AI

To build with AI

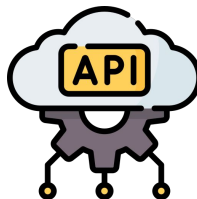
We need model_(LLM)



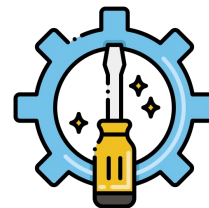
To have model



**Train from
scratch**



Use via API



**Finetune Open
Source**

About Gemma4

#1

The latest generation in Google DeepMind's open model series — Gemma 4, launched in April 2026, built on the Gemini 3 foundation, available in 4 versions ranging from 2B to 31B parameters, under the fully open Apache 2.0 license."

#2

Built on the same technological foundation as Gemini — Gemini is a closed model series, accessed via API/subscription, while Gemma shares the same core technology but can be downloaded and run entirely locally.

#3

The goal is to deliver strong reasoning and coding capabilities while remaining efficient enough to run on commodity hardware — meaning it can run on a personal laptop or a small server, without the need for expensive cloud infrastructure, reducing API costs.



Model variants

Gemma 4 comes in 4 distinct variants.

Parameters	BF16 (16-bit)	SFP8 (8-bit)	Q4_0 (4-bit)
Gemma 4 E2B	9.6 GB	4.6 GB	3.2 GB
Gemma 4 E4B	15 GB	7.5 GB	5 GB
Gemma 4 31B	58.3 GB	30.4 GB	17.4 GB
Gemma 4 26B A4B	48 GB	25 GB	15.6 GB



Model variants

E2B & E4B (Edge models)

Uses an 'Effective parameters' architecture that reduces memory footprint while still achieving high inference performance.

Can run on smartphones and standard laptops — no dedicated GPU required, making it suitable for applications embedded directly on end-user devices.

26B A4B (Workstation model)

Uses a Mixture-of-Experts (MoE) architecture, activating only the necessary 'experts' for each task to optimize resource usage.

Runs on high-end laptops or workstations with GPUs such as the RTX 3090/4090 — thanks to MoE activating only ~4B parameters per inference, it saves significantly on VRAM compared to its actual model size.

31B (Dense model)

A dense model delivering frontier-level intelligence for complex tasks.

Requires a dedicated server or workstation with datacenter GPUs (A100, H100) or at least 2–3 gaming GPUs combined — this is the choice for complex tasks demanding the highest level of accuracy.



Model variants

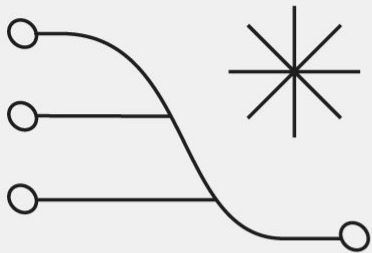
“ Gemma 4 is free to run — but your hardware isn't unlimited. Bigger isn't smarter choice. Every model tier exists for a reason. Know your problem, know your constraints, pick accordingly.

- Gemma4 - ”

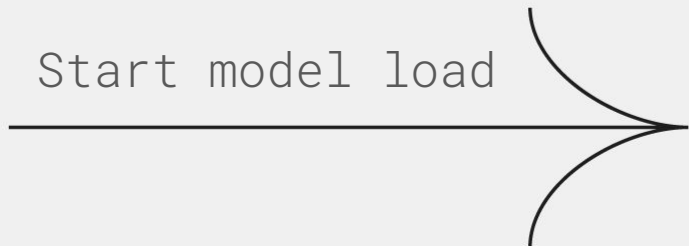




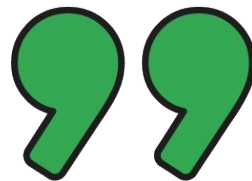
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Start model load



Build  **with AI**



Key features of Gemma4

Thinking mode



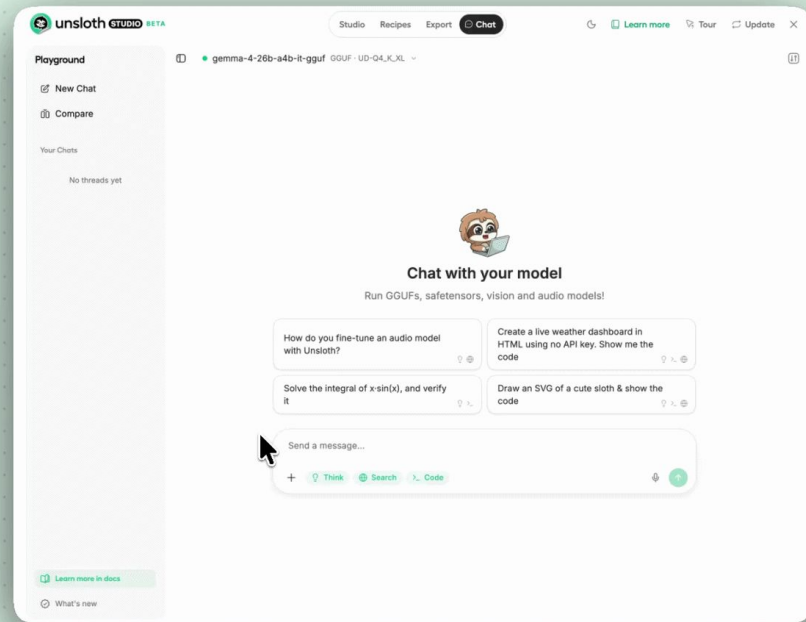
Gemma 4 does not merely predict the next word — it operates as a **reasoning engine**.



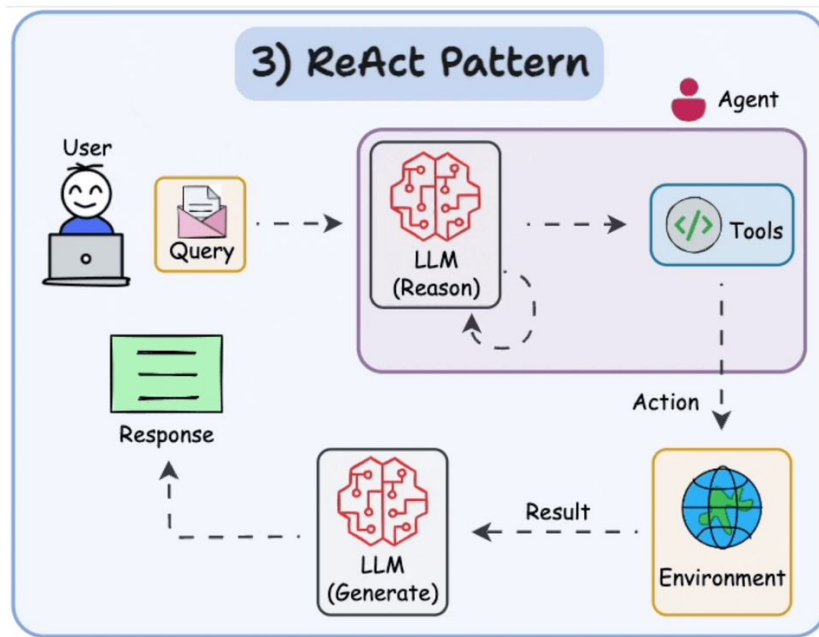
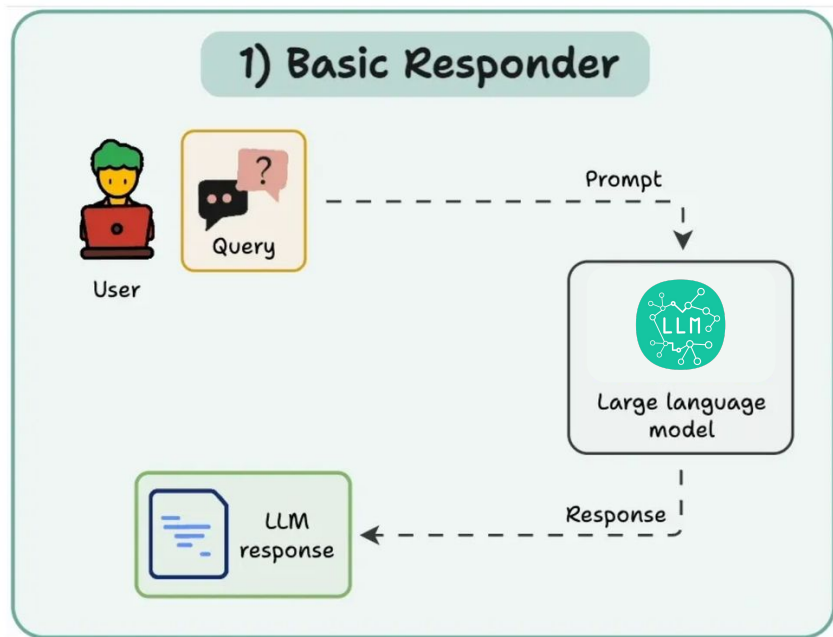
The ability to '**think out loud**' through a hidden or visible scratchpad to work through complex problems before delivering a final answer



Large context window — 256K tokens, enabling the processing of long documents and maintaining multi-turn conversations without losing track of information



ReAct Framework



Extended Multimodality

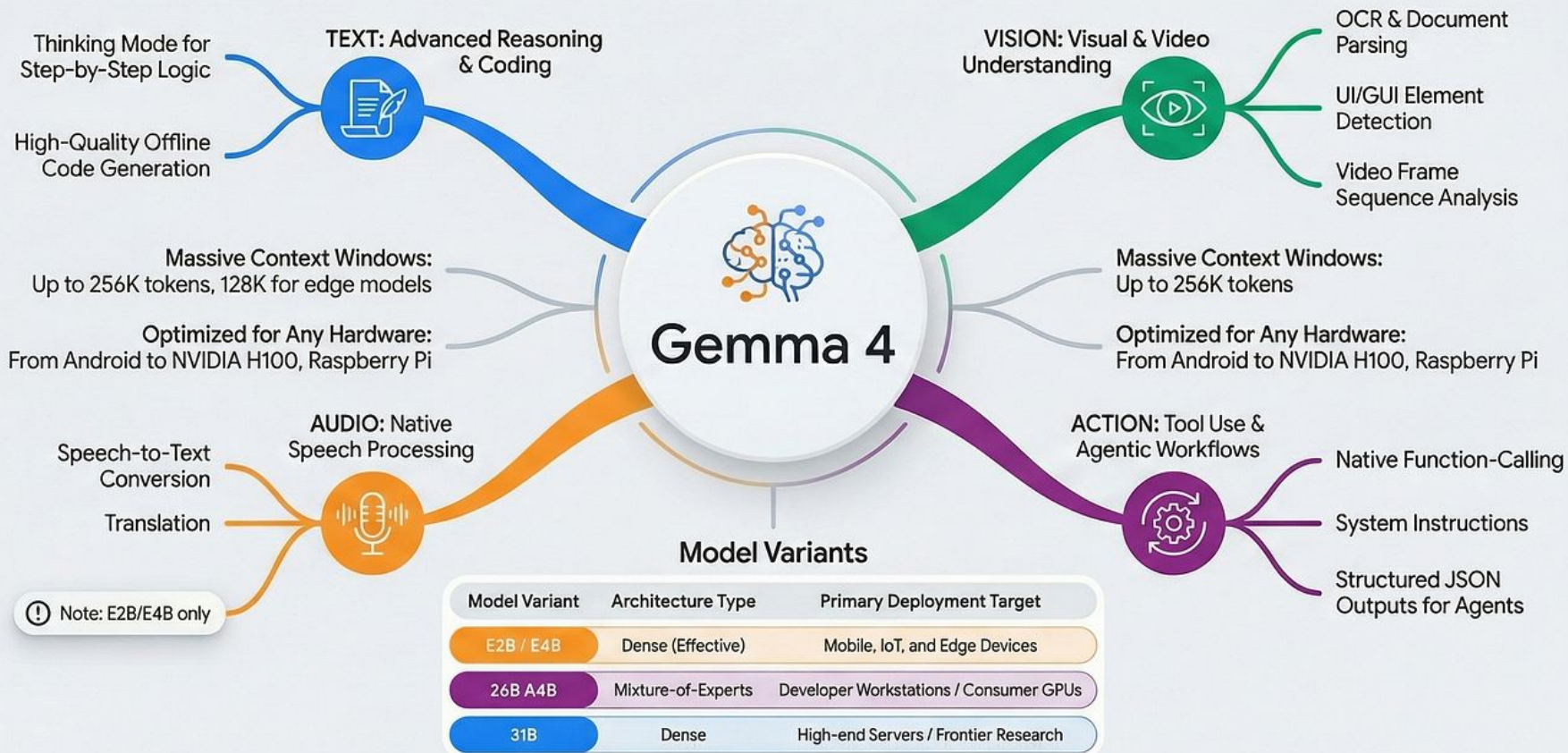


Multimodal Model

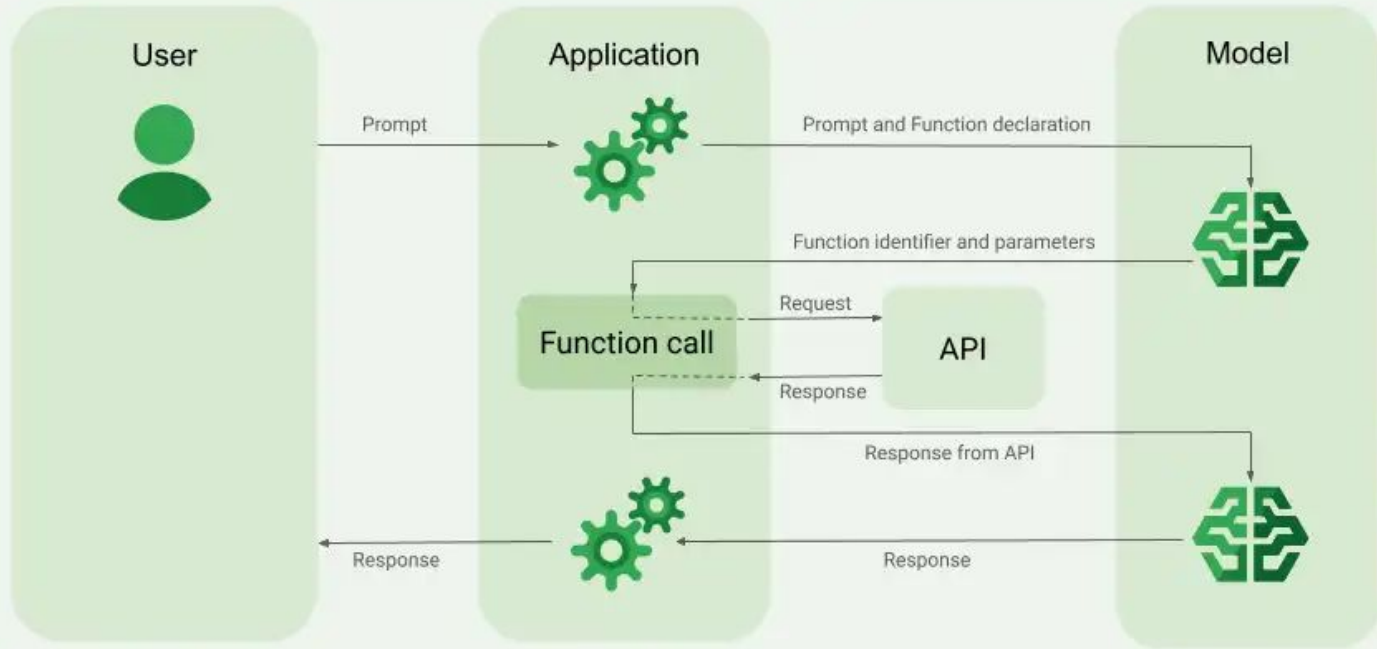
Flexible ability to process inputs in the form of text, images, and audio.



Gemma 4: The Multimodal Capability Map



Function Calling



A blue UFO is positioned at the top of the frame, emitting a bright blue beam of light that tapers downwards. In the center of the beam is a glowing blue gemstone symbol, resembling a four-pointed star or a stylized 'X' within a circle, surrounded by a grid of lines. The background is dark blue with a subtle grid pattern. The text 'Gemma 4' is written in a large, blue, sans-serif font, centered horizontally and partially overlaid by the beam of light.

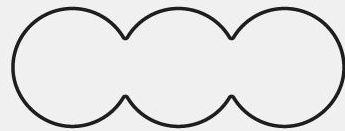
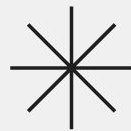
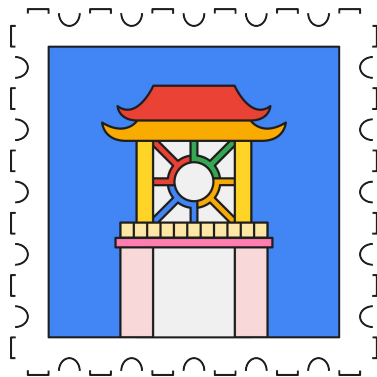
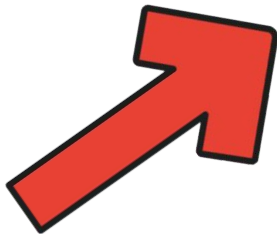
Gemma 4

System Prompt | Function Calling | MCP | Skills



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EcoSystem & Supporting Tools



{ Build  with AI }

Ecosystem and Supporting Tools

Ollama

Run Gemma 4 locally on your own machine — no internet required, no data sent to the cloud. Your queries and outputs stay completely private.



Ollama

Get up and running with large language models.

vLLM

Serve Gemma 4 at scale with high-throughput inference — optimized for speed, efficiency, and production-grade deployment.



Unsloth

Fine-tune Gemma 4 on your own data — faster and more memory-efficient, making customization accessible to everyone.



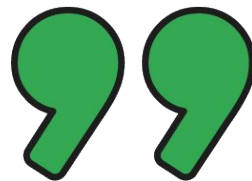
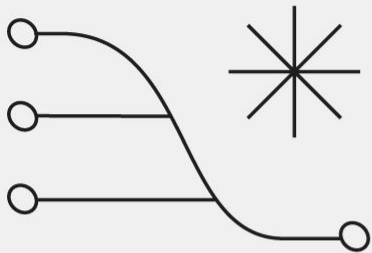
unsloth



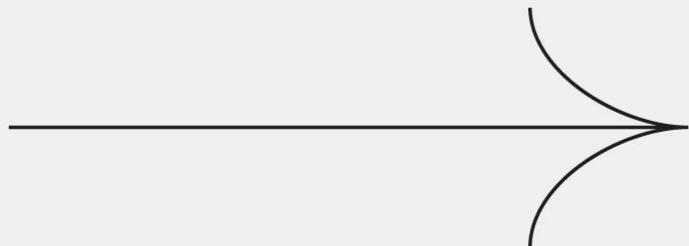
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Demo



Build  with AI

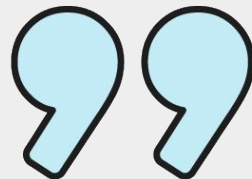
Demo on Collab





Gemma4 - redefined “Open Source AI”

Open source AI doesn't mean “second-rate”.





Thank you!

Google Cloud